Lab Evaluation-02

Course Code: CSE 412

Course Title: Computer Graphics Lab

Experiment Name: Animated Robot with Translation, Rotation and Scaling Using OpenGL

Submitted To:

Khandoker Nosiba Arifin (Lecturer)

Department of CSE

Daffodil International University.

Submitted By

Name: SHARIAR AHAMED RIPON

ID: 0242310005101019

Section: 64_M2

Department of CSE

Daffodil International University.

Submission Date: 16-04-2026

Lab Report: 10

Lab Evaluation-02

Title: Animated Robot with Translation, Rotation and Scaling Using OpenGL

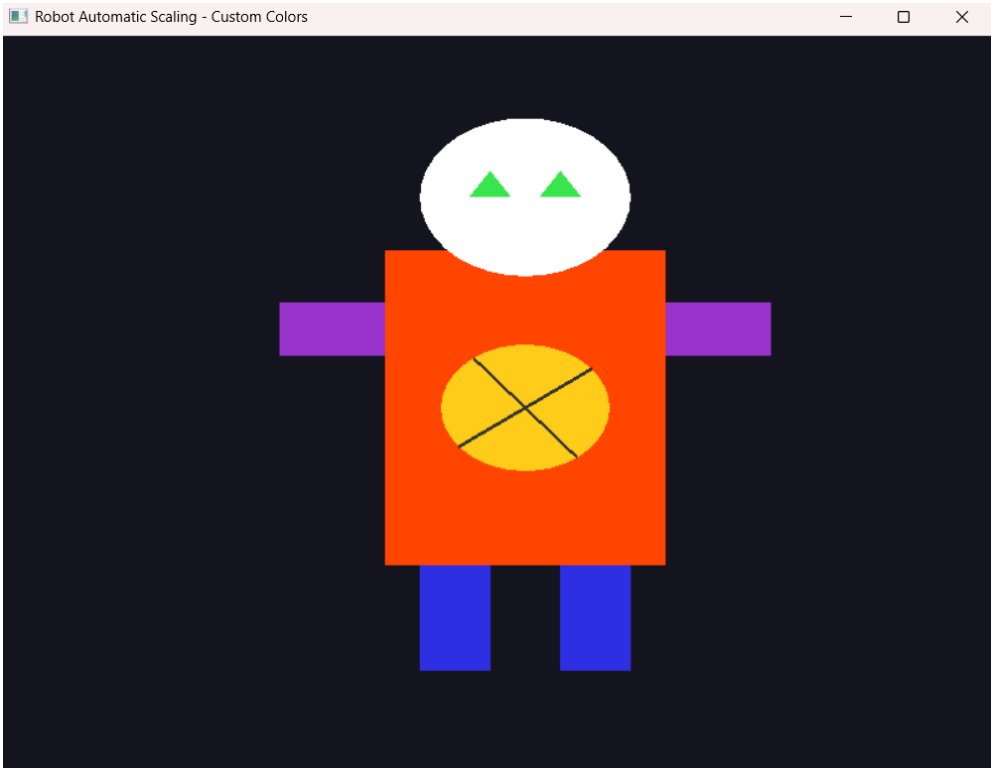
Introduction:

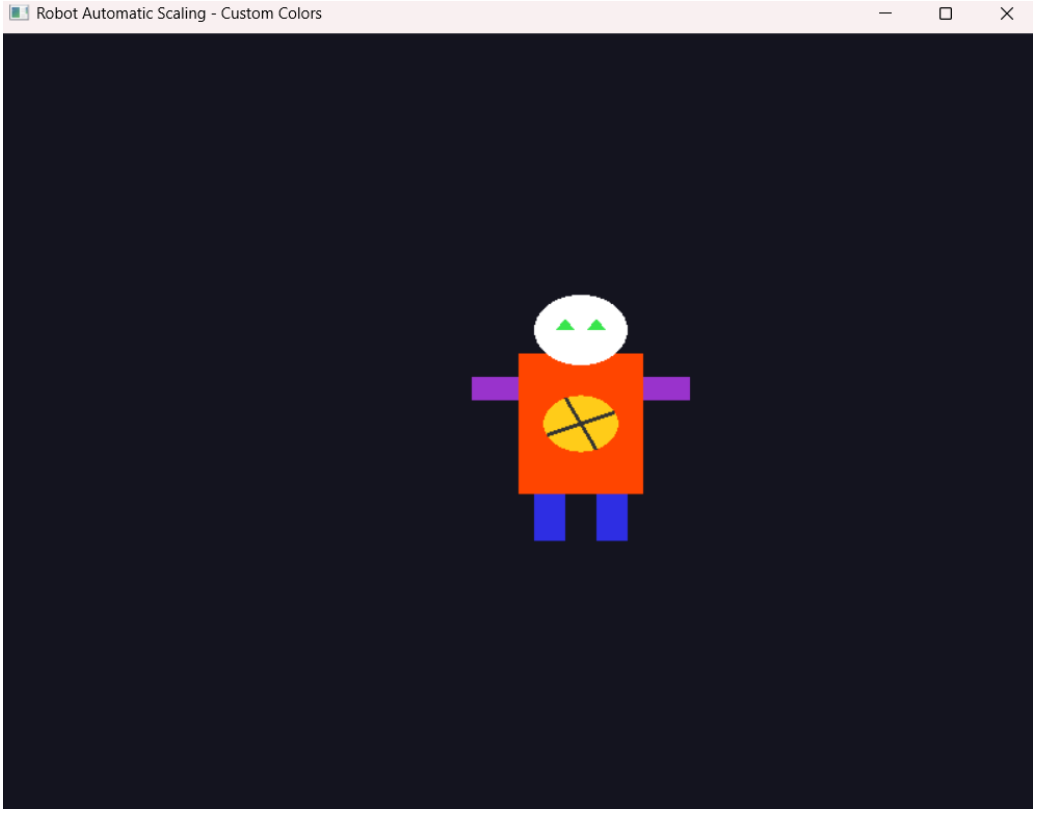
In this project, a robot is created using basic shapes in OpenGL. The robot moves, rotates, and changes its size automatically using animation.

Contents:

- **Functions Used:** `drawCircle()`, `glTranslatef()`, `glScalef()`, `glRotatef()`, `glutTimerFunc()`
- **Concept Used:** Animation, Translation, Rotation, Scaling
- **Shape Used:** Circle, Triangle, Quadrilateral

Graph:





Draw a robot with head, body, arms, and legs. Show movement from left to right, rotation of gear, and size changing effect.

- The full source code of this project is available on GitHub: [Click here]

Main Drawing Code (Important Part):

```
// Translation (Robot Move Left → Right)
glTranslatef(robotX, 0.0f, 0.0f);

// Scaling (Auto Increase/Decrease Size)
glScalef(robotScale, robotScale, 1.0f);

// Robot Body (Rectangle)
glBegin(GL_QUADS);
```

```
    glVertex2f(-0.2f, -0.3f);
    glVertex2f( 0.2f, -0.3f);
    glVertex2f( 0.2f,  0.3f);
    glVertex2f(-0.2f,  0.3f);
glEnd();
// Rotating Gear (Circle)
glPushMatrix();
glRotatef(gearAngle, 0, 0, 1);

glBegin(GL_TRIANGLE_FAN);
for(int i=0;i<=360;i++)
{
    float angle = i * 3.1416 / 180;
    glVertex2f(0.0f + cos(angle)*0.12f,
               0.0f + sin(angle)*0.12f);
}
glEnd();
glPopMatrix();

// Head (Circle)
glBegin(GL_TRIANGLE_FAN);
for(int i=0;i<=360;i++)
{
    float angle = i * 3.1416 / 180;
    glVertex2f(0.0f + cos(angle)*0.15f,
               0.4f + sin(angle)*0.15f);
}
glEnd();
```

Discussion:

The task was to create an animated robot. I used translation for movement, rotation for gear, and scaling to change size automatically using timer.